

FSF3827 Assignment 3

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Question 1

Part 1

Let C be a minimal cover for the knapsack set K . The corresponding valid inequality (cover cut) is:

$$\sum_{i \in C} x_i \leq |C| - 1$$

Let C' be an extension of set C by a single variable, defined as $C' = C \cup \{j\}$ for some $j \notin C$. The cover cut for C' is given by:

$$\begin{aligned} \sum_{i \in C'} x_i &= \sum_{i \in C} x_i + x_j \leq |C'| - 1 \\ &= \sum_{i \in C} x_i + x_j \leq |C| \\ \implies \sum_{i \in C} x_i &\leq |C| - x_j \end{aligned}$$

Since $x_j \in \{0, 1\}$, we observe that:

$$\sum_{i \in C} x_i \leq |C| - 1 \leq |C| - x_j$$

Thus, the cover cut for C dominates the cover cut for C' because it imposes a more restrictive bound. Consequently, the cover cut for C' is weaker.

Part 2

Lets lift a variable $x_k \notin C'$ into the cover cut for C . The lifted inequality is given by:

$$\sum_{i \in C} x_i + \alpha_{k_1} x_k \leq |C| - 1$$

Where the lifting coefficient α_{k_1} is defined as:

$$\alpha_{k_1} = |C| - 1 - \max \left(\sum_{i \in C} x_i \mid x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup k \right)$$

Similarly, we lift the same variable x_k into the cover cut for C' . The lifted inequality can thus be expressed as:

$$\sum_{i \in C'} x_i + \alpha_{k_2} x_k \leq |C'| - 1 \implies \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j$$

Here the lifting coefficient α_{k_2} is defined as:

$$\begin{aligned}
\alpha_{k_2} &= |C'| - 1 - \max \left(\sum_{i \in C'} x_i \mid x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C' \cup k \right) \\
&= |C| - \max \left(\sum_{i \in C} x_i + x_j \mid x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup j \cup k \right) \\
&\geq |C| - \max x_j - \max \left(\sum_{i \in C} x_i \mid x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup k \right) \\
&\geq |C| - 1 - \max \left(\sum_{i \in C} x_i \mid x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup k \right) \\
&\geq \alpha_{k_1}
\end{aligned}$$

Thus we can see that:

$$\begin{aligned}
&\sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j \\
\Rightarrow \sum_{i \in C} x_i + \alpha_{k_1} x_k &\leq \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j \\
\Rightarrow \sum_{i \in C} x_i + \alpha_{k_1} x_k &\leq \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - 1 \leq |C| - x_j
\end{aligned}$$

Therefore, the lifted cover cut for C dominates the lifted cover cut for C' , making the latter weaker.

Question 2

Question 3

Part 2

Let us lift a variable $x_k \notin C'$ into the cover cut for C . The lifted inequality is given by:

$$\sum_{i \in C} x_i + \alpha_{k_1} x_k \leq |C| - 1$$

where the lifting coefficient α_{k_1} is defined as:

$$\alpha_{k_1} = |C| - 1 - \max \left\{ \sum_{i \in C} x_i : x_k = 1, x \in \{0, 1\}^n, \sum_{i \in N} a_i x_i \leq b, N = C \cup \{k\} \right\}$$

Similarly, we lift the same variable x_k into the cover cut for C' . The lifted inequality is:

$$\sum_{i \in C'} x_i + \alpha_{k_2} x_k \leq |C'| - 1 \Rightarrow \sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j$$

Here, the lifting coefficient α_{k_2} is defined as:

$$\begin{aligned}
\alpha_{k_2} &= |C'| - 1 - \max \left\{ \sum_{i \in C'} x_i : x_k = 1, \sum_{i \in C' \cup \{k\}} a_i x_i \leq b \right\} \\
&= |C| - \max \left\{ \sum_{i \in C} x_i + x_j : x_k = 1, \sum_{i \in C \cup \{j, k\}} a_i x_i \leq b \right\} \\
&\geq |C| - \max(x_j) - \max \left\{ \sum_{i \in C} x_i : x_k = 1, \sum_{i \in C \cup \{k\}} a_i x_i \leq b \right\} \\
&\geq |C| - 1 - \max \left\{ \sum_{i \in C} x_i : x_k = 1, \sum_{i \in C \cup \{k\}} a_i x_i \leq b \right\} \\
&\geq \alpha_{k_1}
\end{aligned}$$

Since $\alpha_{k_2} \geq \alpha_{k_1}$, we can compare the inequalities. Starting with the lifted minimal cover cut:

$$\sum_{i \in C} x_i + \alpha_{k_1} x_k \leq |C| - 1$$

Since $x_j \leq 1$, we know that $|C| - 1 \leq |C| - x_j$. Thus:

$$\sum_{i \in C} x_i + \alpha_{k_1} x_k \leq |C| - x_j$$

The lifted extended cover cut requires $\sum_{i \in C} x_i + \alpha_{k_2} x_k \leq |C| - x_j$. Since the minimal cover cut implies the bound $|C| - x_j$ using a smaller (or equal) coefficient α_{k_1} , it is more restrictive than the extended cover cut. Thus, the extended cut is weaker.

Question 4

Question 5